Street Mapper: Automated Pothole Detection using YOLO-based Deep Learning

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Abstract

Road infrastructure monitoring is critical for ensuring transportation safety and efficiency. Traditional methods of pothole detection rely heavily on manual inspection, which is time-consuming, expensive, and often inconsistent. In this project, we propose an automated pothole detection system using a deep learning-based object detection model. Leveraging the YOLO (You Only Look Once) architecture, the system is trained to detect potholes in images and video streams in real time. The proposed approach demonstrates promising performance in identifying road damages and can be integrated into smart city systems for proactive maintenance and monitoring.

1. Introduction

Maintaining road quality is a significant challenge faced by municipalities worldwide. Potholes not only cause vehicle damage but also pose serious safety risks. Manual inspection methods are inefficient and fail to provide real-time insights.

With advancements in computer vision and deep learning, automated detection systems have become feasible. Object detection models, particularly those in the YOLO family, provide real-time detection capabilities with high accuracy. This project aims to develop a system that can:

- Detect potholes in images and video streams
- Provide real-time alerts with confidence scores
- Serve as a foundation for smart road monitoring systems

2. Related Work

Recent research in road damage detection has utilized deep learning techniques such as Convolutional Neural Networks (CNNs) and object detection frameworks like Faster R-CNN, SSD, and YOLO.

- **Faster R-CNN** provides high accuracy but is computationally expensive
- **SSD (Single Shot Detector)** improves speed but may compromise accuracy
- **YOLO** offers a balance between speed and accuracy, making it suitable for real-time applications

Studies have shown that YOLO-based models outperform traditional methods in real-time detection scenarios, especially in edge-device deployments.

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3. Methodology

3.1 Dataset Preparation

The dataset consists of labeled images of potholes. Each image is annotated with bounding boxes and class labels in YOLO format.

The dataset is divided into:

- Training set
- Validation set

Directory structure:

```
datasets/
  └── potholes/
      ├── train/
      └── valid/
```

3.2 Data Configuration

A YAML configuration file is used to define dataset paths and class information:

- Number of classes: 1
- Class name: pothole

This configuration is used by the training pipeline to load and process data efficiently.

3.3 Model Architecture

The project uses a pre-trained YOLO model (YOLO11s variant). YOLO is a single-stage object detection model that predicts bounding boxes and class probabilities simultaneously.

Key advantages:

- Real-time detection capability
- High inference speed
- End-to-end training

3.4 Training Process

The model is fine-tuned using transfer learning on the pothole dataset.

Training parameters:

- Epochs: 80
- Image size: 640 × 640
- Batch size: 16
- Device: GPU (T4)

The training process optimizes the model to detect potholes with high confidence while minimizing false positives.

3.5 Detection and Alert System

After training, the model is deployed for inference on images and video streams.

A simulated alert system is implemented:

- Detects pothole
- Outputs confidence score
- Triggers location-based alert

Example output:

```
DAMAGE DETECTED
Location: Latitude, Longitude
Confidence: 0.85
```

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4. Experimental Results

4.1 Performance Evaluation

The model demonstrates effective detection of potholes across various scenarios, including:

- Different lighting conditions
- Varying road textures
- Partial occlusions

Performance depends heavily on dataset quality and diversity.

4.2 Observations

- High confidence detections in clear road conditions
- Reduced accuracy in low-light or blurred images
- False positives in cases of shadows or road patches

4.3 Limitations

- Limited dataset size affects generalization
- No real-time GPS integration (simulated only)
- Performance may drop in extreme weather conditions

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5. Discussion

The results indicate that YOLO-based models are highly effective for real-time pothole detection. The trade-off between speed and accuracy is well-balanced, making the system suitable for deployment in real-world scenarios.

However, improvements are needed in:

- Dataset diversity
- Robustness to environmental variations
- Integration with real-time data pipelines

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6. Conclusion

This project presents a deep learning-based approach to pothole detection using the YOLO framework. The system successfully detects road damages and demonstrates the feasibility of automated infrastructure monitoring.

The proposed solution can significantly reduce manual effort and enable proactive maintenance in smart cities.

7. Future Work

Future enhancements may include:

- Integration with GPS for real-time location tracking
- Deployment on mobile or edge devices
- Use of larger and more diverse datasets
- Integration with government dashboards for reporting
- Multi-class detection (cracks, bumps, debris)

References

1. Redmon et al., “You Only Look Once: Unified, Real-Time Object Detection”
2. Ultralytics YOLO Documentation
3. Recent research papers on road damage detection using deep learning

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